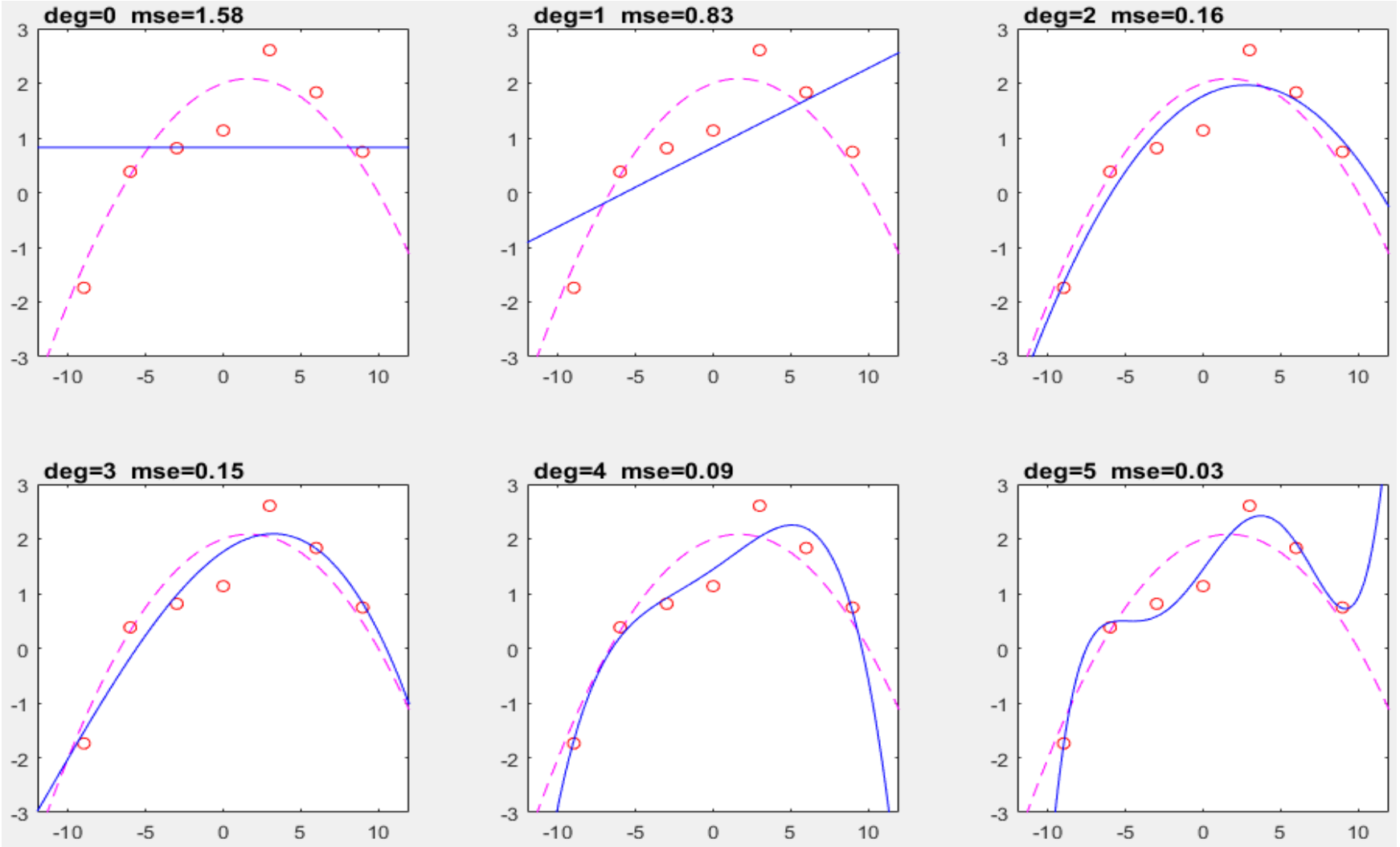
# Feature Space Geometry (2)

# Dimensionality and Representative Power

- **Representational power** refers to the capacity of a mathematical model or feature space to capture, distinguish, and express complex patterns and decision boundaries present in data.
- As feature dimensionality increases, representational power grows because higher-dimensional spaces provide more degrees of freedom to separate overlapping classes and resolve fine-grained data structures.

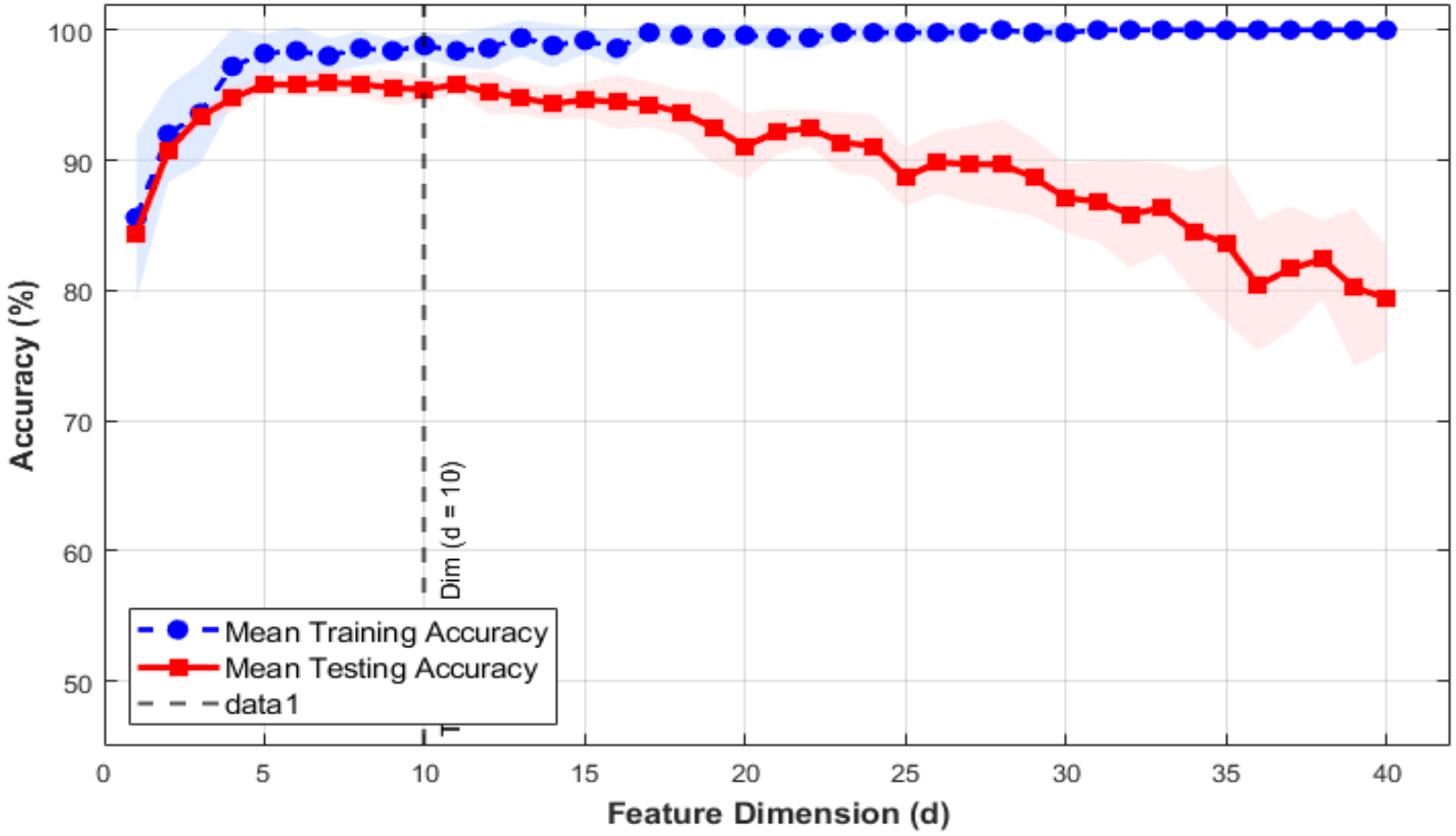
# Dimensionality and Representative Power

Polynomial regression as an example.
(Each coefficient as a degree of freedom.)



# The Hughes Phenomenon

- As representation power (dimensionality) increases, the test accuracy peaks at some dimensionality and starts to drop.
- A demo using Mahalanobis distance based classification (two classes, 10 true dimensions):



# Bias-Variance Dilemma

Desired properties of the model:

- Low **Bias** (goodness of fitting):
  - The estimated mapping should fit to the training data as well as possible.
  - Goal: To capture the behaviors of the underlying mapping as well as possible.
- Low **Variance** (high confidence of prediction):
  - Assume that we have several estimated mappings, each derived from a different set of training samples.
  - For a given $\boldsymbol{x}$, each mapping gives an estimation for $\boldsymbol{y}$.
  - The estimations for $\boldsymbol{y}$ should by as similar to each other as possible.
  - Goal: To be confident of the estimation of $\boldsymbol{y}$.

# Bias-Variance Dilemma

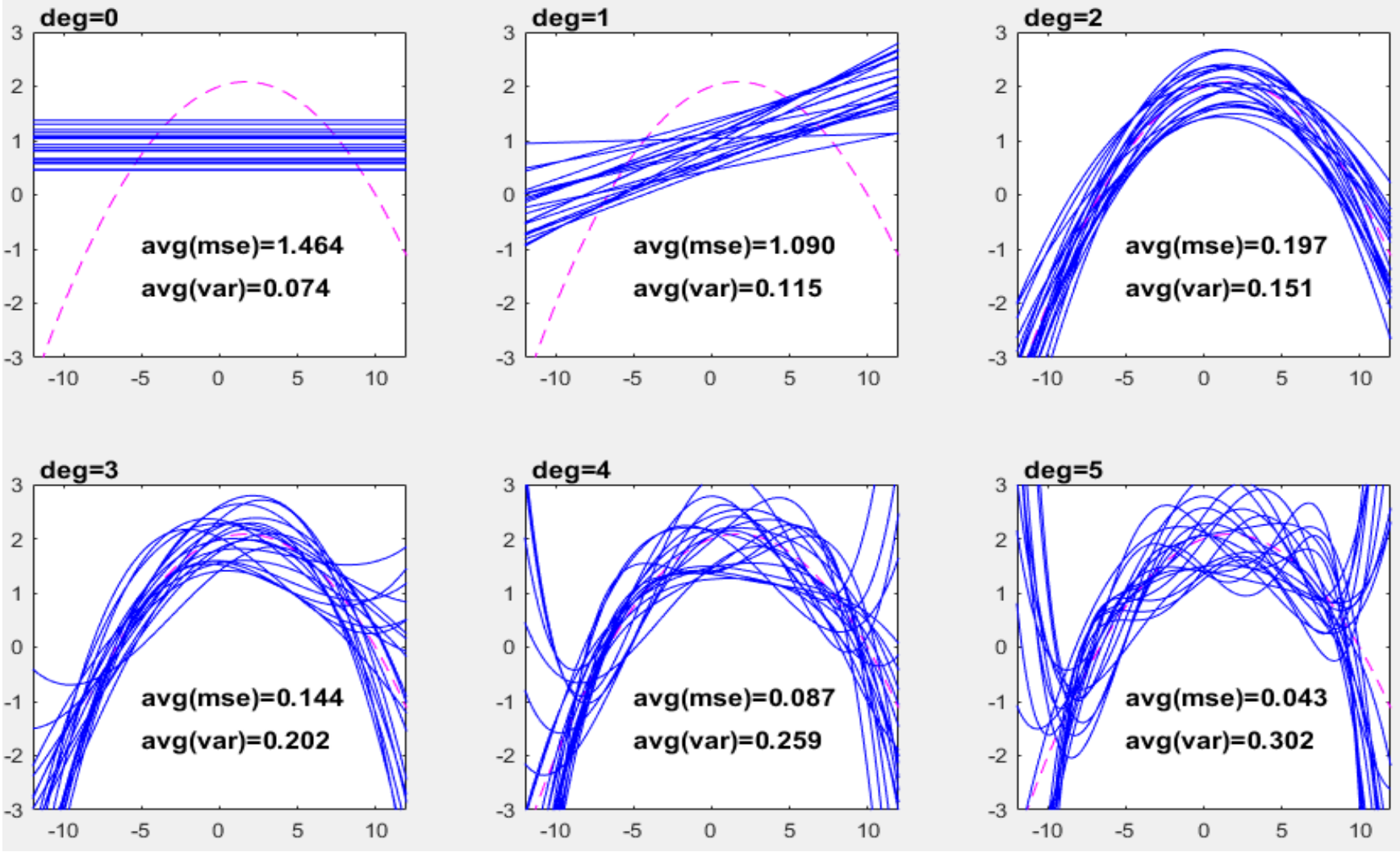
Now, difficulties for supervised learning:

- To reduce **Bias**
  - ➔ Use more complex models
  - ➔ BOTH signal and noise are learned
  - ➔ Overfitting
- To reduce **Variance**
  - ➔ Try to avoid learning noise (variance is caused by noise)
  - ➔ Use less complex models
  - ➔ Signal is not sufficiently learned
  - ➔ Underfitting

Here we use the term "noise" to represent variations in data that are not relevant to the prediction.

# Illustration of Variance

Polynomial regression as an example.
(Each coefficient as a degree of freedom.)




Each degree is sampled 20 times.

# Bias-Variance Dilemma

- The components of expected prediction error:

$$\mathbb{E}_{\mathcal{D}}\left[(y-\hat{f}(x))^2\right] = \underbrace{\mathrm{Bias}^2\left(\hat{f}(x)\right)}_{\text{structural underfitting}} + \underbrace{\mathrm{Variance}\left(\hat{f}(x)\right)}_{\text{parameter estimation noise}} + \underbrace{\sigma_\epsilon^2}_{\text{irreducible noise}}$$

- Effect on different classifiers:
  - Mahalanobis distance, Gaussian maximum-likelihood: Strong (due to the covariance matrix estimation)
  - Nonparametric method (e.g., kNN): Caused by distance concentration at high dimensionality
  - Linear classifiers with regularization or extra constraint (e.g., SVM): Less serious (can better handle small-sample-size problems)

# What to Do Now with Variance?

Several representative approaches:

- Increasing the number of training samples.
  - Data augmentation, resampling, …
- Dimensionality reduction (e.g., PCA)
- Consensus-based methods (ensembles)
  - Examples include random forest, dropout during training, etc.
- Regularization or other structural priors
  - The term "regularization" can refer to approaches that induce some desired properties of a model that are not the primary objective.
- Trying to determine the suitable level of model complexity (feature dimensionality)
  - Normally done with validation

# Regularization Example (L2 Regularization)

- Include the **bias-variance** trade-off during the learning process:
  - **Regularization**: Modify the learning objective to minimize BOTH the bias and the model complexity. (Normally, the learning process only attempts to minimize bias.)
  - Example **cost function** (to be minimized) for polynomial regression:

To minimize:

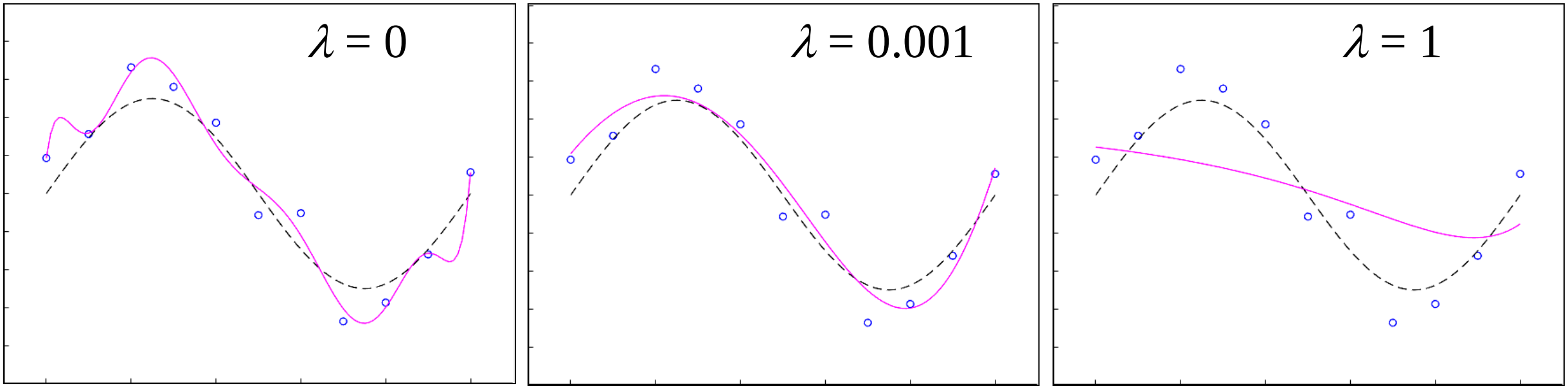
$$E = \sum_i \left| y_i - \sum_{k=0}^{d} a_k x_i^k \right|^2 + \lambda \sum_{k=0}^{d} a_k^2$$

Fitting Error (reducing bias)

Regularization (reducing complexity)

# Regularization Example (L2 Regularization)

- Example (polynomial regression):




$N$=11, degree=9

# Ridge Regularization (for Covariance Matrix)

- Rationale (the high-dimensional challenge): When feature dimension approaches or exceeds sample size, empirical sample covariance becomes ill-conditioned or singular, causing matrix inversion to amplify sampling noise and degrade model generalization.
- Principle (Shrinkage): Ridge regularization stabilizes sample covariance by adding a spherical penalty term to the covariance matrix:

$$\hat{\Sigma}_{\text{ridge}} = \hat{\Sigma} + \gamma I \quad (\gamma > 0)$$

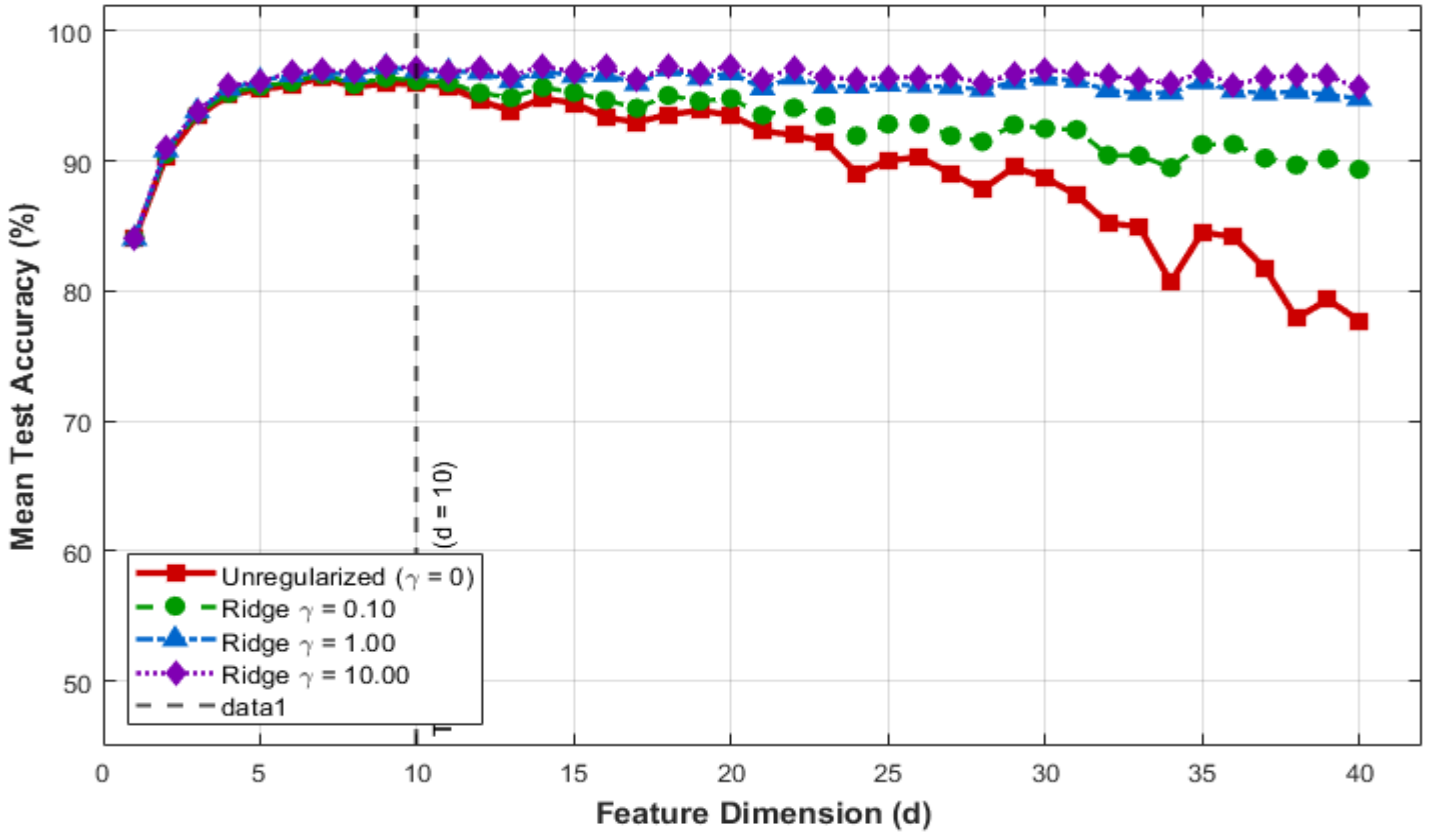
- Geometric Mechanism: Moving the minimum eigenvalue away from zero, reducing the eccentricity of the covariance ellipsoid to prevent noise amplification along low-variance (noise) axes.

# Ridge Regularization (for Covariance Matrix)

- The Bias-Variance Tradeoff: Intentionally introduces a controlled amount of model bias in exchange for a reduction in parameter estimation variance.
- Spatial whitening with ridge regularization (PCA as example):

$$x_{\text{white}} = (\Sigma + \gamma I)^{-1/2} x.$$

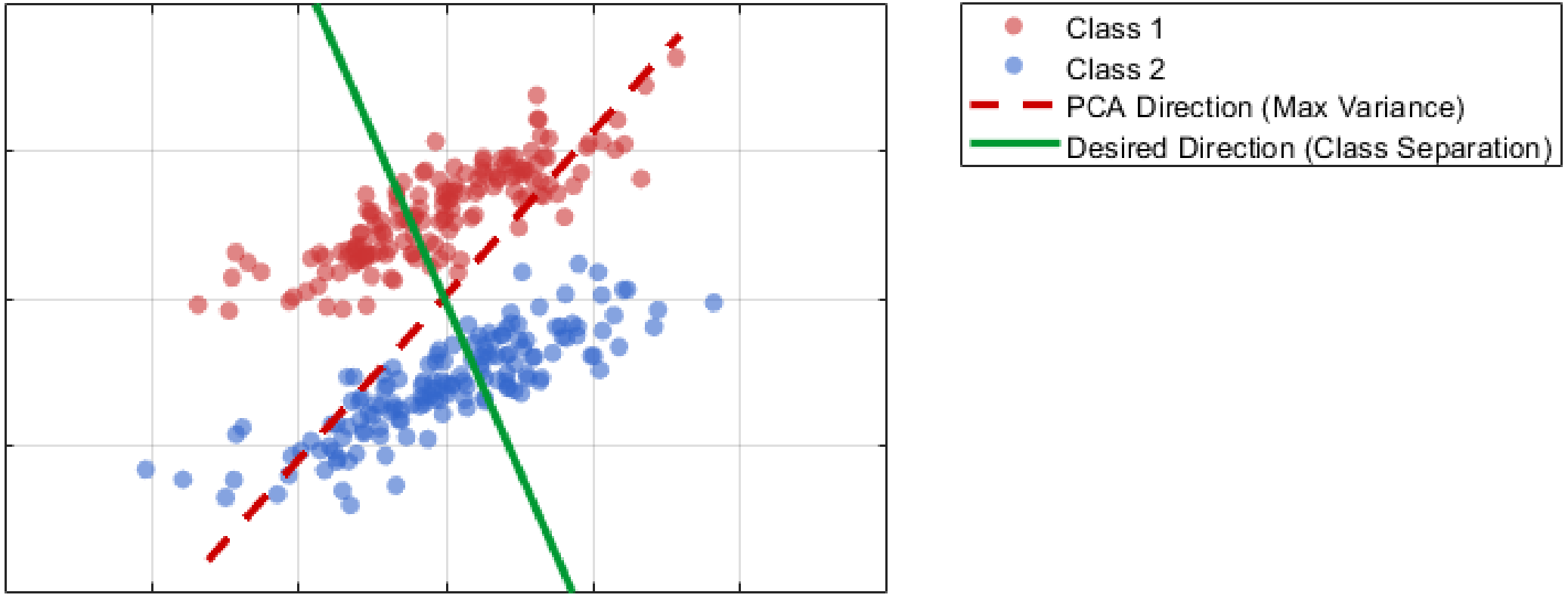
# Ridge Regularization and Hughes Phenomenon




- Overfitting (high variance) occurs much slower with ridge regularization.
- In general, there is a small added bias (loss of peak accuracy) with high regularization.
- The optimal weight $\gamma$ is generally determined via validation.

# Supervised Dimensionality Reduction: Motivation

Unsupervised methods (e.g., PCA) may generate projections that cause classes distributions to overlap, thus degrading subsequent classification results.

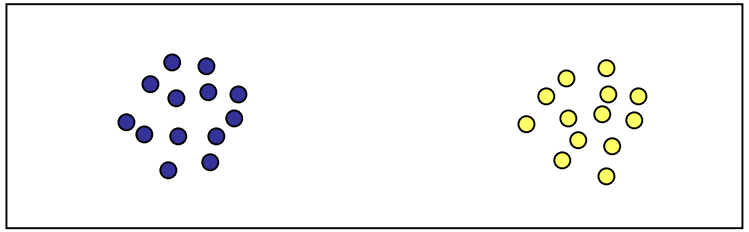


# Class Separability

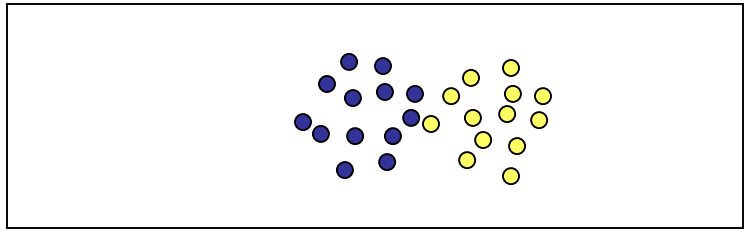
When the target task is classification, we want the distribution of samples in the reduced-dimensionality space to be suitable for classification. Example criterion:

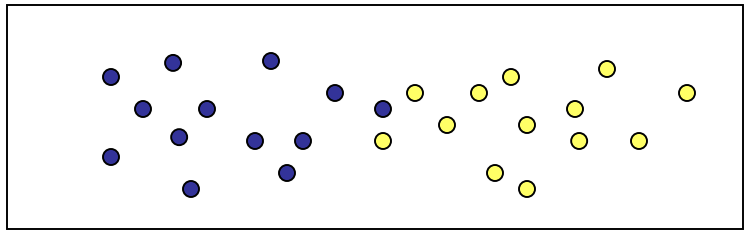
**Class separability**: How well the training samples of different classes are separated from each other.

Higher separability

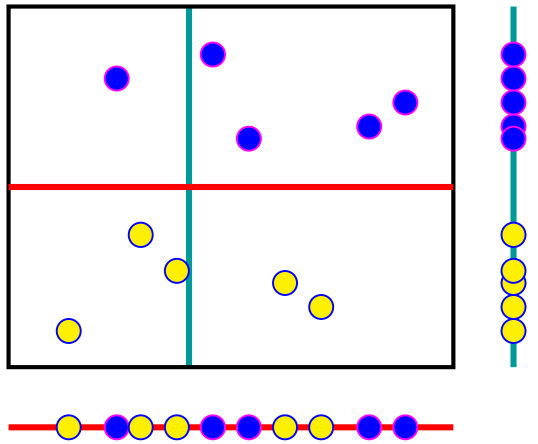


Lower separability





(Projection to 1D)



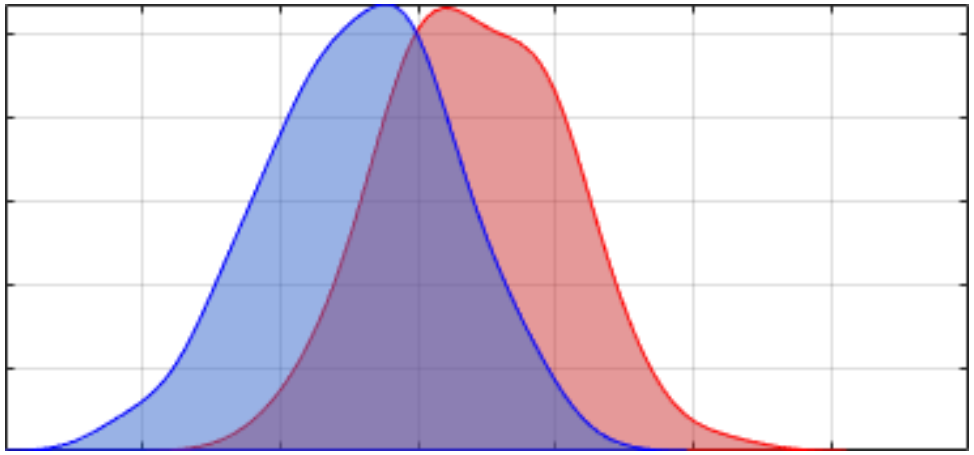
Higher separability

Lower separability

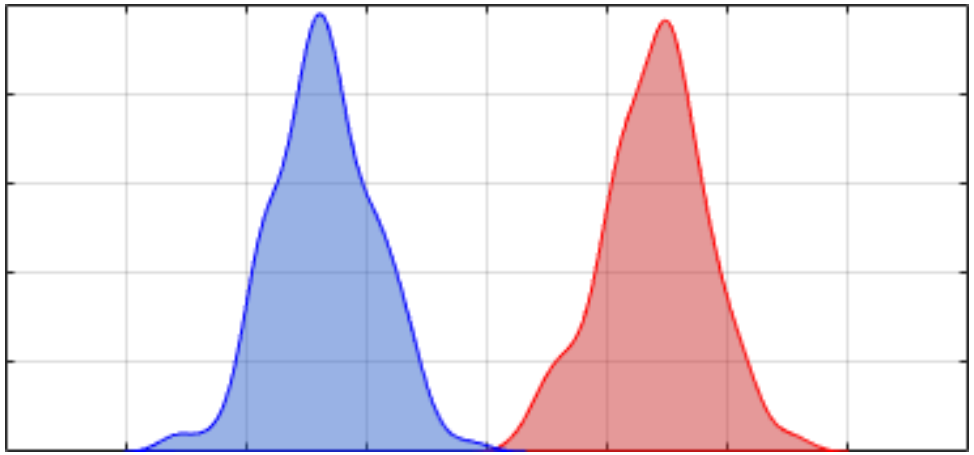
# Class Separability




Lower separability (PCA)



Higher separability



# Scatter Matrices

Class separability is commonly computed using scatter matrices. Here we consider:

- Within-class scatter matrix: $S_w = \sum_{i=1}^{M} P_i \Sigma_i$
- Between-class scatter matrix: $S_b = \sum_{i=1}^{M} P_i (\boldsymbol{\mu}_i - \boldsymbol{\mu}_0)(\boldsymbol{\mu}_i - \boldsymbol{\mu}_0)^T$
- Mixture scatter matrix: $S_m = E\left[(\mathbf{x} - \boldsymbol{\mu}_0)(\mathbf{x} - \boldsymbol{\mu}_0)^T\right]$

It can be proved that

$$S_m = S_w + S_b$$

$M$: # classes
$P_i$: per-class probability
$\Sigma_i$: per-class covariance
$\boldsymbol{\mu}_i$: per-class mean
$\boldsymbol{\mu}_0$: global mean

# Scatter Matrices and Class Separability Measures

The following class separability measures are defined based on the scatter matrices:

$$J_1 = \frac{\mathrm{tr}\{S_m\}}{\mathrm{tr}\{S_w\}} \qquad J_2 = \frac{|S_m|}{|S_w|} = \left|S_w^{-1} S_m\right| \qquad J_3 = \mathrm{tr}\left\{S_w^{-1} S_m\right\}$$

Fisher's discriminant ratio: A class separability measure in 1-D, equiprobable cases:

$$FDR = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} \propto \frac{S_b}{S_w}$$

In multi-class cases, we can use the average over all pairs of classes.

FDR can be useful in quantifying the class separability achievable with a single feature or a 1-D projection.

# Feature Subset Selection

The goal is to find the best $l$ features that retain as much information for classification in the original $m$ features as possible.

Choosing the best $l$ individual features

- Pro: Simple and fast
- Con: Loss of information of the correlation between features

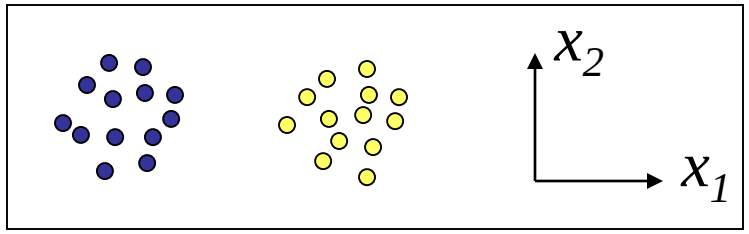
Choosing the best $l$-feature subset

- Pro: Retains the information of the correlation between features
- Con: Impractical to find the "optimal" subset (too many combinations to test)

# Scalar Feature Selection

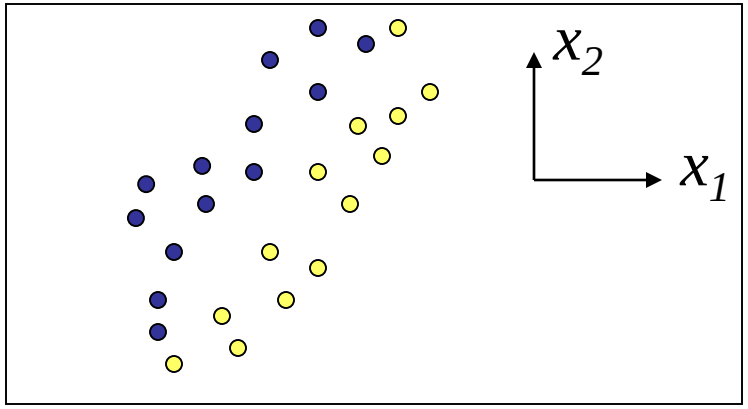
The idea is to test each of the $m$ available features individually, and select the best $l$ of them according to some criterion, such as a class separability measure.

Example:




Feature $x_1$ is better than feature $x_2$.

Problem with correlated features:




Neither feature is good individually, but they are useful when used together.

# Sequential Feature Selection

Greedy approaches for (sub-optimal) feature subset selection:

Let $X_k$ be the $k$-element feature subset selected.
Let $C$ be the criterion used.
The goal is to choose $l$ of $m$ features.

**Sequential Forward Selection**

Start with $X_0 = \{\}$
For $k = 1$ to $l$
  For all $x \notin X_{k-1}$, compute $C$ for $\{x\} \cup X_{k-1}$
  $X_k \leftarrow$ the $\{x\} \cup X_{k-1}$ that gives the best $C$

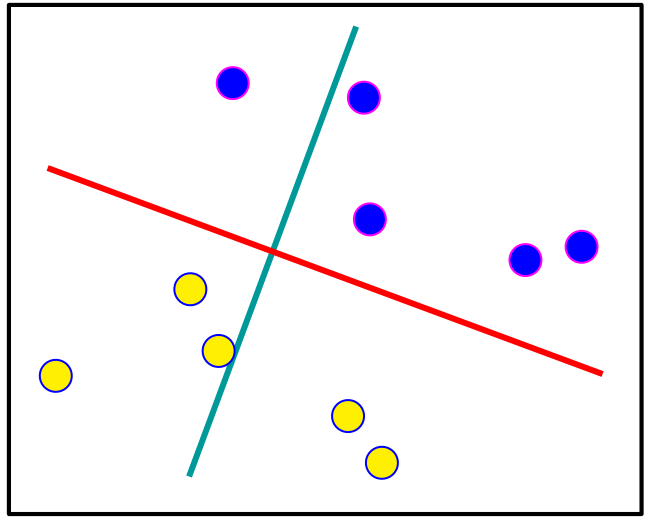
**Sequential Backward Selection**

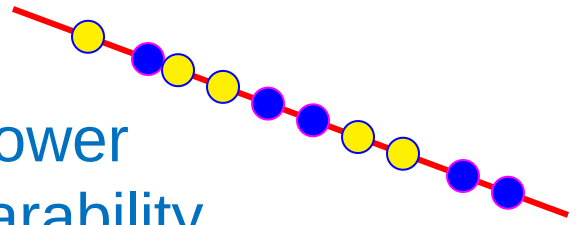
Start with $X_l = \{\text{all } m \text{ features}\}$
For $k = m-1$ to $l$ (decreasing)
  For all $x \in X_{k+1}$, compute $C$ for $X_{k+1} \backslash \{x\}$
  $X_k \leftarrow$ the $X_{k+1} \backslash \{x\}$ that gives the best $C$

# Fisher's Linear Discriminant (FLD)

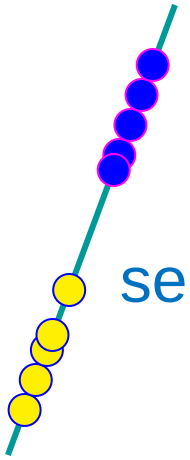
Goal: To maximize a class separability measure while projecting the feature vectors onto a one-dimensional subspace of the original $m$-dimensional feature space.

Example 2-D, 2-class case when projected to 1-D:





Lower separability



Higher separability

# Fisher's Linear Discriminant

Let us start by finding the projection $\boldsymbol{w}$ that optimizes **FDR**: $y = \boldsymbol{w}^T \boldsymbol{x}$

For each class, the mean and variance are also transformed similarly: $\mu_{iy} = \boldsymbol{w}^T \boldsymbol{\mu}_{ix}$ and $\sigma_{iy}^2 = \boldsymbol{w}^T \Sigma_{ix} \boldsymbol{w}$

For 2-class cases, FDR becomes $FDR(\boldsymbol{w}) = S_{wy}^{-1} S_{by} = (\boldsymbol{w}^T S_{wx} \boldsymbol{w})^{-1} (\boldsymbol{w}^T S_{bx} \boldsymbol{w})$

To maximize it, we set its partial derivative with respect to $\boldsymbol{w}$ to zero, which leads to

$$S_{bx}\boldsymbol{w} = \left( \frac{\boldsymbol{w}^T S_{bx} \boldsymbol{w}}{\boldsymbol{w}^T S_{wx} \boldsymbol{w}} \right) S_{wx}\boldsymbol{w} \Rightarrow S_{bx}\boldsymbol{w} = \lambda S_{wx}\boldsymbol{w}$$

A generalized eigenvalue problem

If $S_{wx}$ is invertible, we get $S_{wx}^{-1} S_{bx} \boldsymbol{w} = \lambda \boldsymbol{w}$

The vector $S_{bx}\boldsymbol{w}$ is always along the $(\boldsymbol{\mu}_{1x} - \boldsymbol{\mu}_{2x})$ direction: $S_{bx}\boldsymbol{w} \propto (\boldsymbol{\mu}_{1x} - \boldsymbol{\mu}_{2x})$

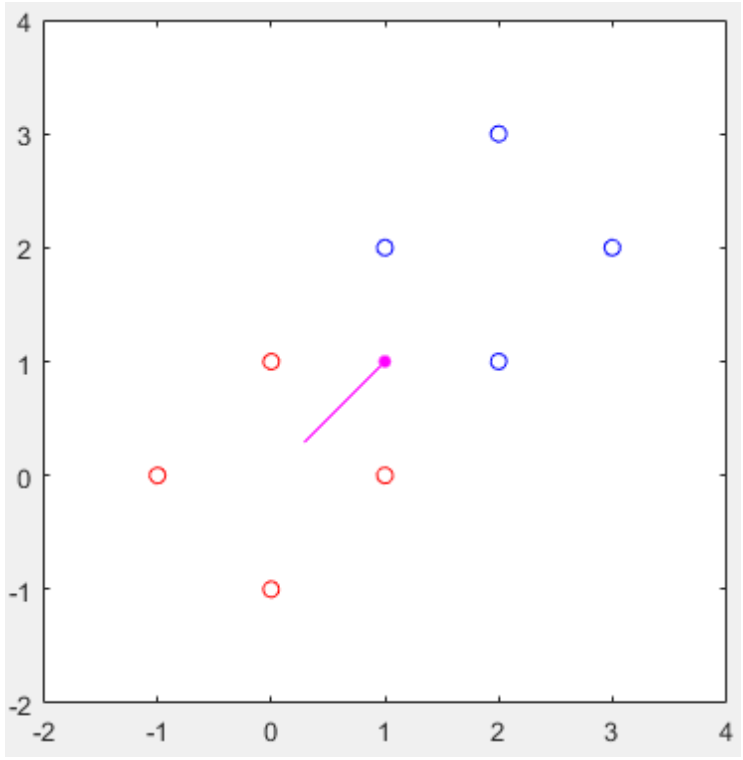
We can then obtain $\boldsymbol{w}$ (ignoring scalar factors) as $\boldsymbol{w} = S_{wx}^{-1}(\boldsymbol{\mu}_{1x} - \boldsymbol{\mu}_{2x})$

# FLD and Examples

- FLD, when projection to 1-D, can be used as a linear classifier with a separately specified threshold. It can also be directly evaluated using, say, ROC curves.
- For problems of $M$ classes, the maximum dimensions of projection is $M$–1 using FLD (sometimes called Linear Discriminant Analysis (LDA)).

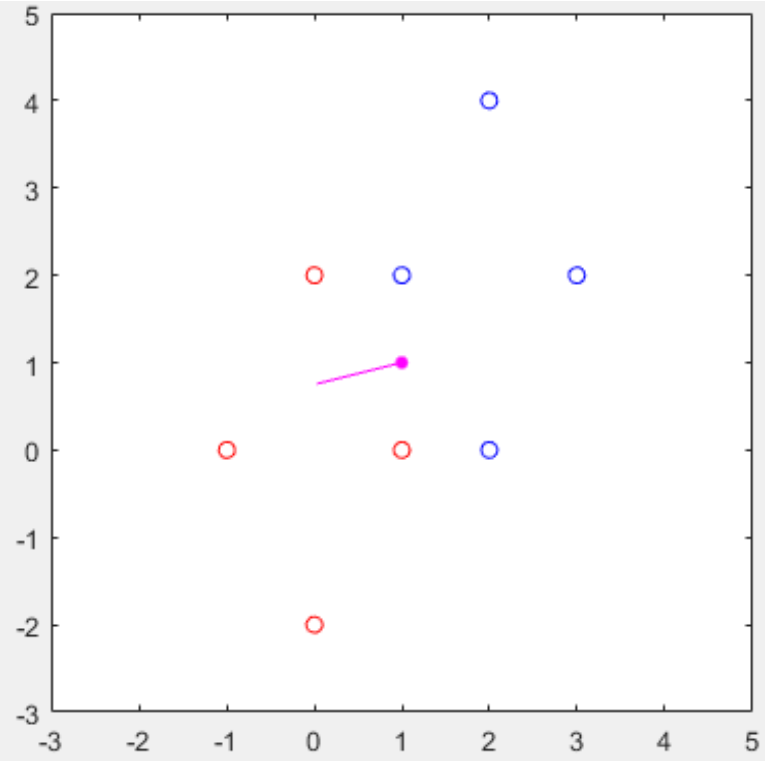
$$X_1 = \begin{bmatrix} 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 2 & 2 & 3 & 1 \\ 1 & 3 & 2 & 2 \end{bmatrix}$$




$$X_1 = \begin{bmatrix} 0 & 0 & 1 & -1 \\ 2 & -2 & 0 & 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 2 & 2 & 3 & 1 \\ 0 & 4 & 2 & 2 \end{bmatrix}$$



# Linear Discriminant Analysis

Goal: To maximize a class separability measure while projecting the feature vectors onto a $l$-dimensional subspace of the original $m$-dimensional feature space.

The class separability measure used: $$J = \mathrm{tr}\left\{ S_w^{-1} S_b \right\}$$

Let $A$ be the matrix representing the projection: $$\boldsymbol{y} = A^T \boldsymbol{x}$$

Transformed scatter matrices: $$S_{yw} = A^T S_{xw} A \quad \text{and} \quad S_{yb} = A^T S_{xb} A$$

Regularized LDA: Apply ridge regularization to $\boldsymbol{S_w}$ (particularly useful for small-sample-size problems.

# Linear Discriminant Analysis

The condition $\frac{\partial J_y}{\partial A} = 0$ now yields $S_{xw}^{-1} S_{xb} A = A S_{yw}^{-1} S_{yb}$

Using a matrix $B$ that simultaneously diagonalizes $S_{yw}$ and $S_{yb}$:

$B^T S_{yw} B = I$ and $B^T S_{yb} B = D$ ($D$ is diagonal)

We obtain $(S_{xw}^{-1} S_{xb}) C = CD$ where $C=AB$.

This again is an eigenvalue problem. (We end up with $C$ while looking for $A$; this is ok because the value of $J$ is the same using either one as the transformation matrix.)

# Linear Discriminant Analysis

The rank of $S_{xb}$ is limited to $M-1$ (i.e., at most $M-1$ non-zero eigenvalues). So is $S_{xw}^{-1}S_{xb}$. As a result, we can project the data onto a subspace of at most $M-1$ dimensions, i.e., $l \leq M-1$. The special case $M=2$ gives FLD.

Let us now express $J_y$, which we want to maximize, using the sum of eigenvalues of $S_{xw}^{-1}S_{xb}$:

$$J_y = \mathrm{tr}\{D\}$$

When using $l < M-1$, we discard some eigenvectors of $S_{xw}^{-1}S_{xb}$ from $C$ and their corresponding eigenvalues from $D$. The value of $J_y$ is then the sum of the remaining eigenvalues. So we maximize $J_y$ by projecting to the $l$-dimensional subspace containing the eigenvectors associated with the $l$ largest eigenvalues.

# Fisherfaces vs. Eigenfaces

- Eigenfaces: PCA based with no consideration for the intended classification task.
- Fisherfaces: LDA followed by PCA:
  - PCA first: Original dimensionality → $N - C$ dimensions
  - LDA next: $N - C$ dimensions → $C - 1$ dimensions
- A principle useful for small-sample-size problems.